Inspiring Excellence

Department of Electrical and Electronic Engineering

Hardware / Software(✔) Report-3 Submission

Semester: Spring 26
Course Code: EEE283L
Course Title: Digital Logic Design
Section:03

Experiment No:06

Experiment Name: Implementation of Counters

Group No: 02

Group members:

Sl.	***Name***	***ID***
2.	Faiyaz Hasin Reza	24121374

Date of Submission: 29/03/26

Prepared by:

Faiyaz Hasin Reza

Experiment Name: Implementation of Counter

Introduction:

Counter circuits are essential components in digital systems used to count events, generate timing intervals, and track elapsed time. In digital electronics, counters are mainly two types: Asynchronous and Synchronous. Asynchronous counters are simpler in design but may have output delays because their flip-flops are triggered by different clock signals. In contrast, synchronous counters have all flip-flops triggered by the same clock pulse, ensuring coordinated operation. This experiment focuses on using the 74161 IC, a 4-bit synchronous binary counter, to implement both a standard binary counter and a BCD counter.

Objective:

The objective of this experiment is to build and verify a 4-bit synchronous binary counter and a BCD counter using the 74161 IC. By using the Proteus software, the goal is to observe how coordinated clock pulses and external NAND gates can control specific counting sequences, such as resetting the count at ten for BCD operation.

Software:

- Proteus & Professional
- Digital logic library components inside Proteus

Components:

- Subcircuit component
- Logic toggle
- LED indicators
- 4 bit counter(74161)
- Connecting wires
- NAND gate

Simulation Diagram:

4 bit counter

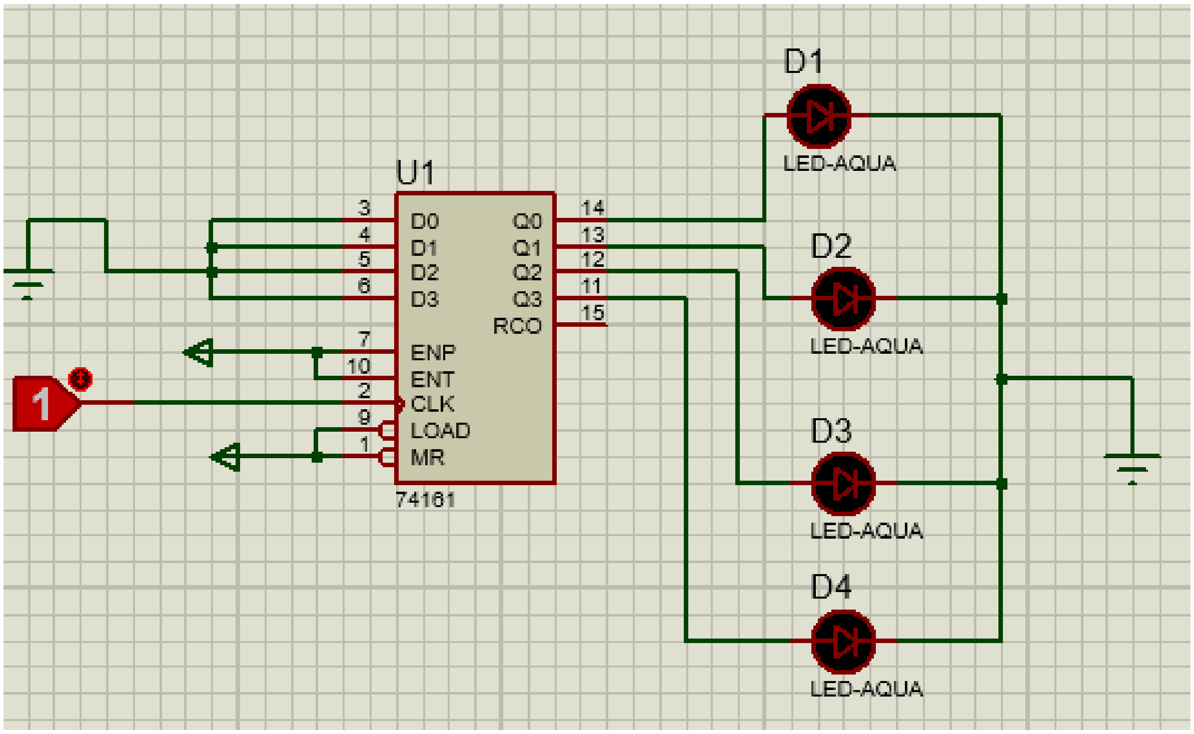

Figure:- 4 bit counter circuit diagram

Truth table (4 bit counter)

Clock pulse	D4 (LED)	D3 (LED)	D2 (LED)	D1 (LED)	Decimal
Initial	0	0	0	0	0
1	0	0	0	1	1
2	0	0	1	0	2
3	0	0	1	1	3
4	0	1	0	0	4
5	0	1	0	1	5
6	0	1	1	0	6
7	0	1	1	1	7
8	1	0	0	0	8
9	1	0	0	1	9
10	1	0	1	0	10
11	1	0	1	1	11
12	1	1	0	0	12
13	1	1	0	1	13
14	1	1	1	0	14
15	1	1	1	1	15

Test Result:-

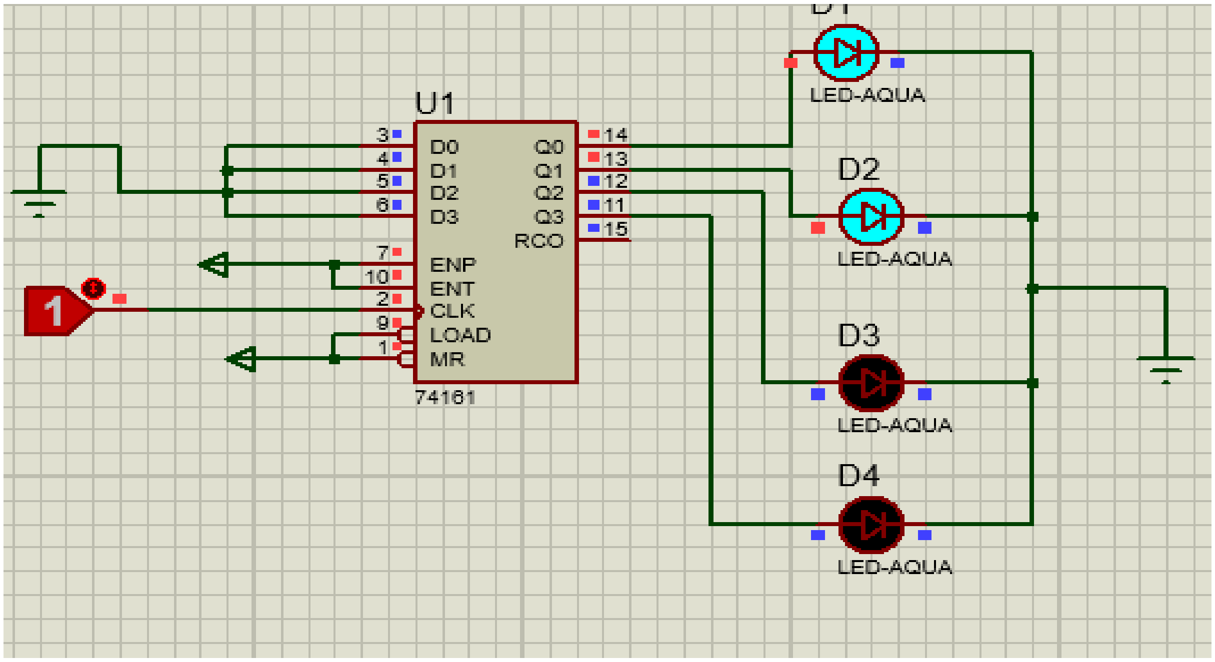

Figure:-Simulation showing the 4-bit counter at 0011_2(Decimal 3) with all output LEDs in the OFF state.

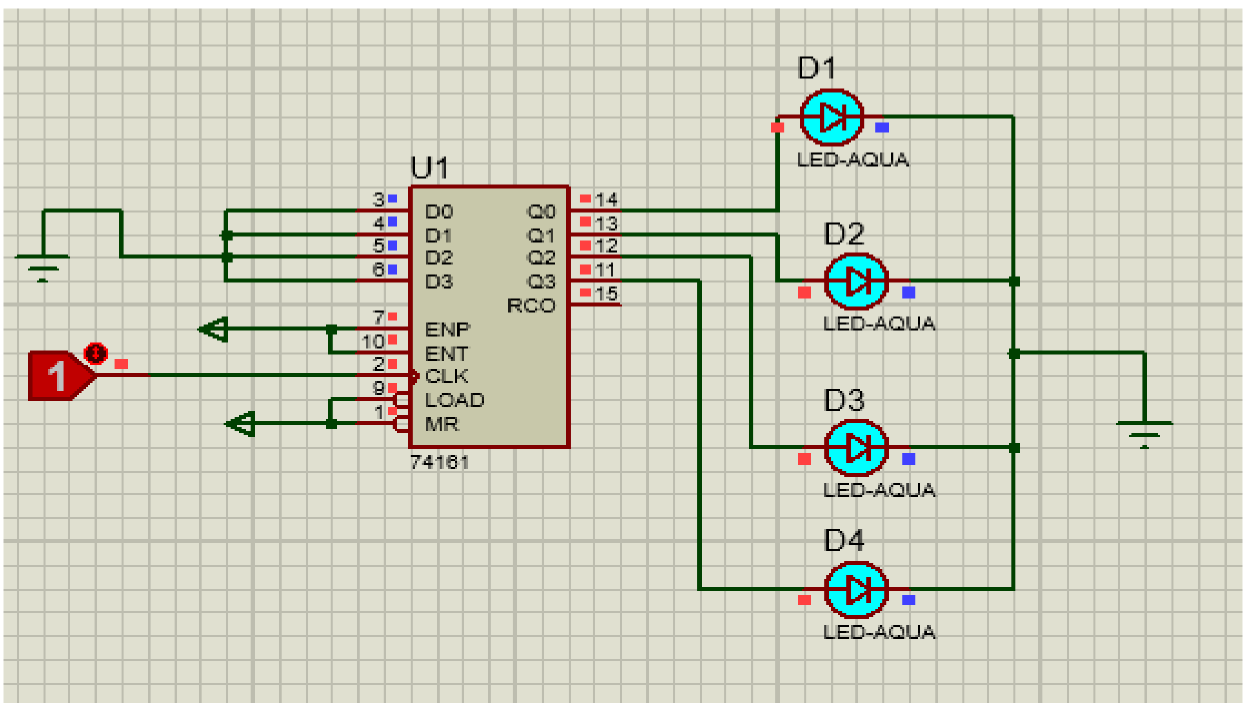

Figure:-Final simulation state showing the 4-bit counter at 1111_2 (Decimal 15) with all output LEDs in the ON state.

4 bit BCD Counter

Simulation Diagram:

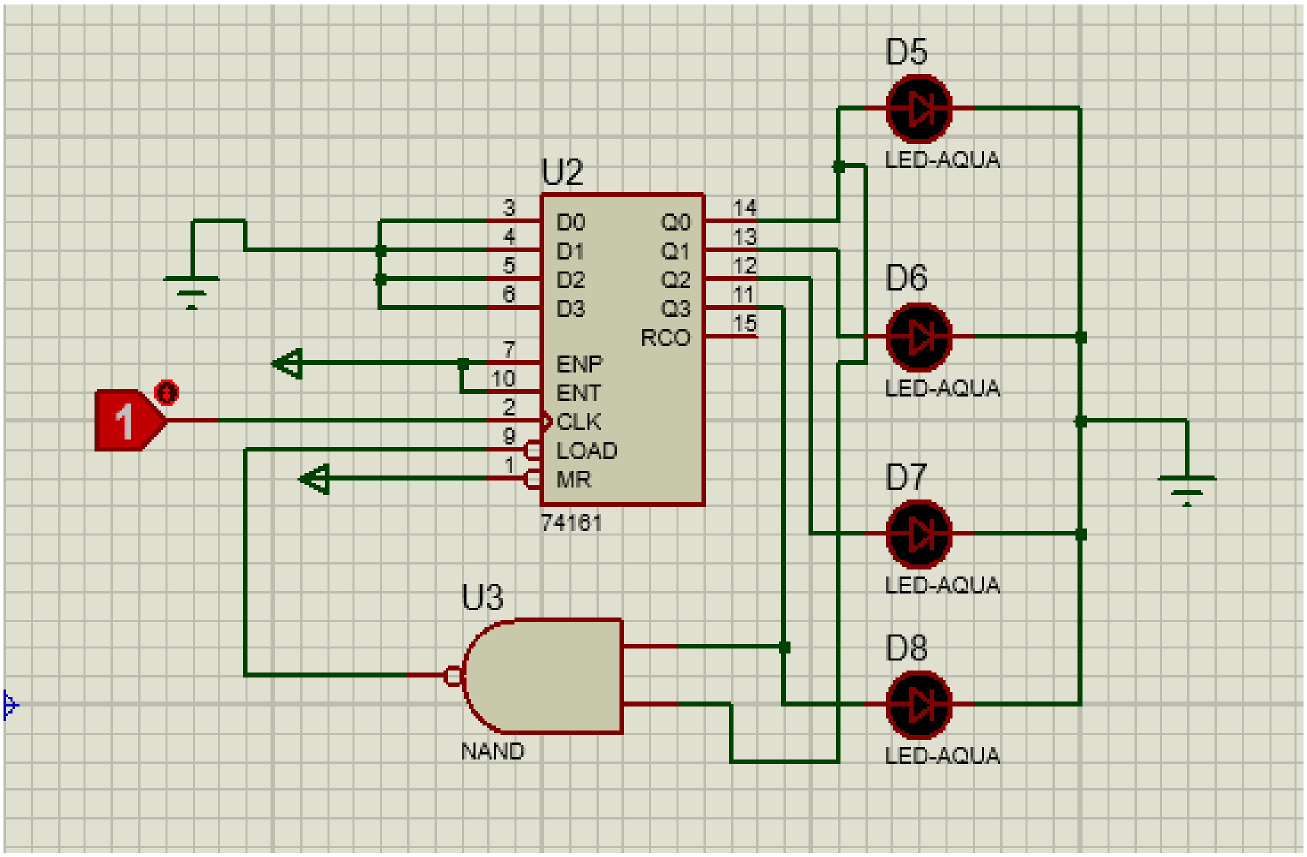

Figure:- 4 bit BCD counter circuit diagram

Truth table (4 bit BCD counter)

Clock pulse	D8 (LED)	D7 (LED)	D6 (LED)	D5 (LED)	Decimal
Initial	0	0	0	0	0
1	0	0	0	1	1
2	0	0	1	0	2
3	0	0	1	1	3
4	0	1	0	0	4
5	0	1	0	1	5
6	0	1	1	0	6
7	0	1	1	1	7
8	1	0	0	0	8
9	1	0	0	1	9
10(reset)	0	0	0	0	0

Test Result:-

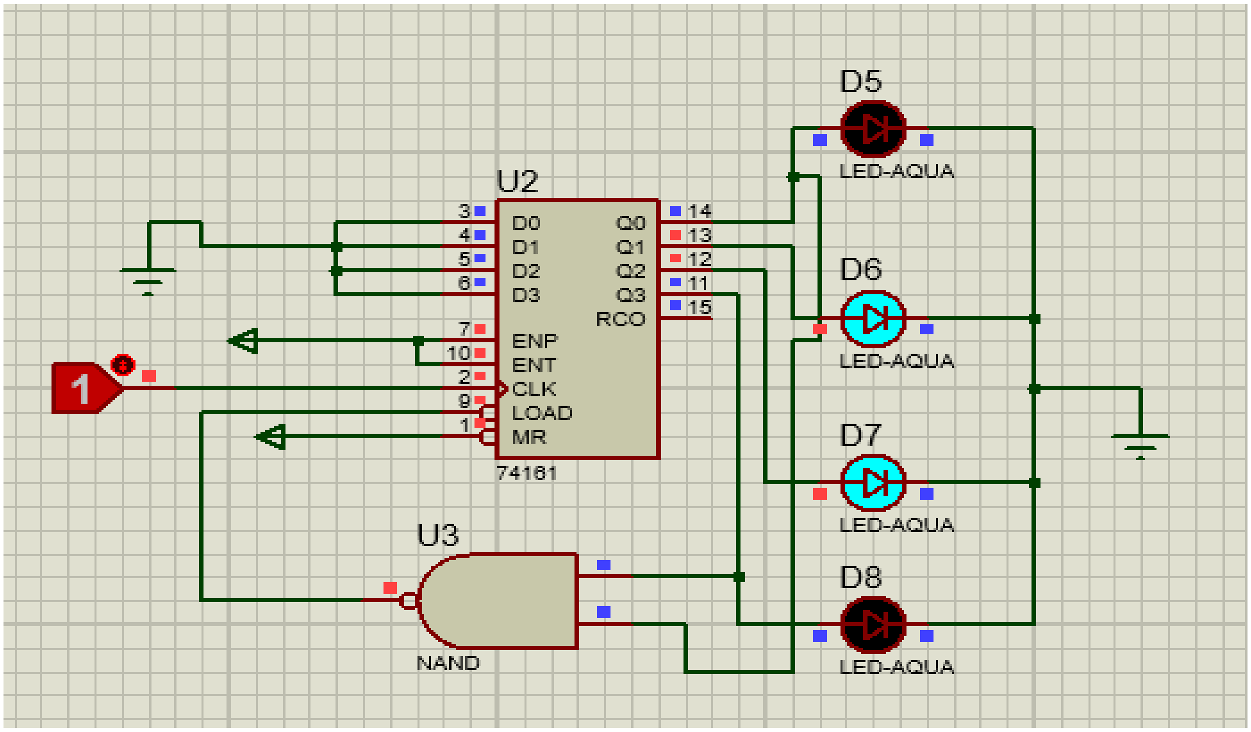

Figure:-Simulation showing the 4-bit BCD counter at 0110_2(Decimal 6) with all output LEDs in the OFF state.

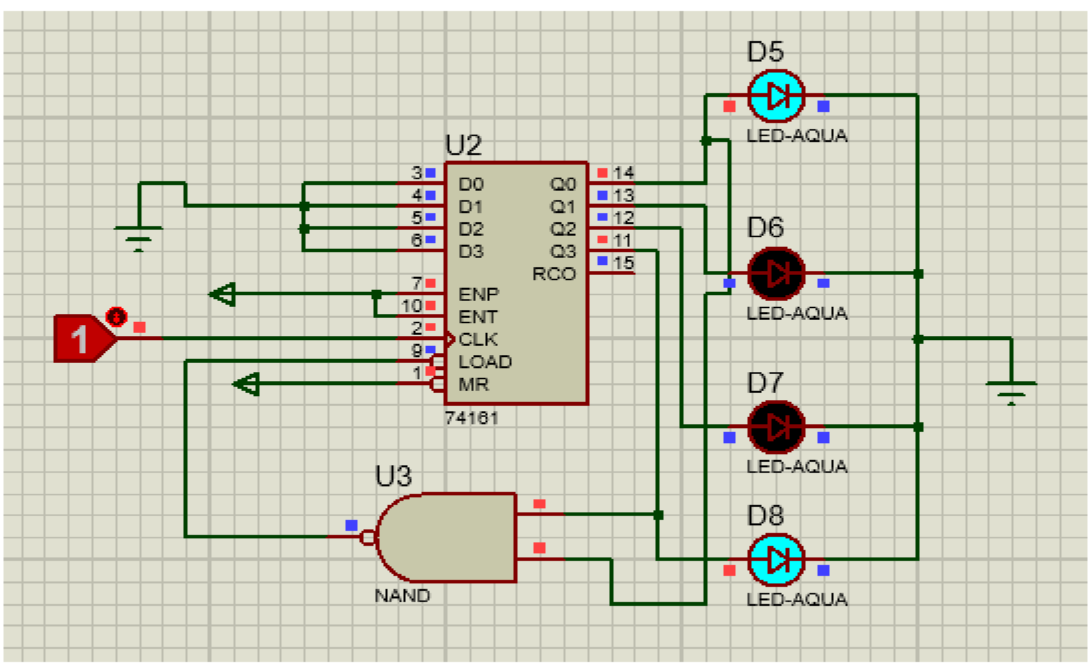

Figure:-Simulation showing the 4-bit BCD counter at 1001_2(Decimal 9) with all output LEDs in the OFF state.

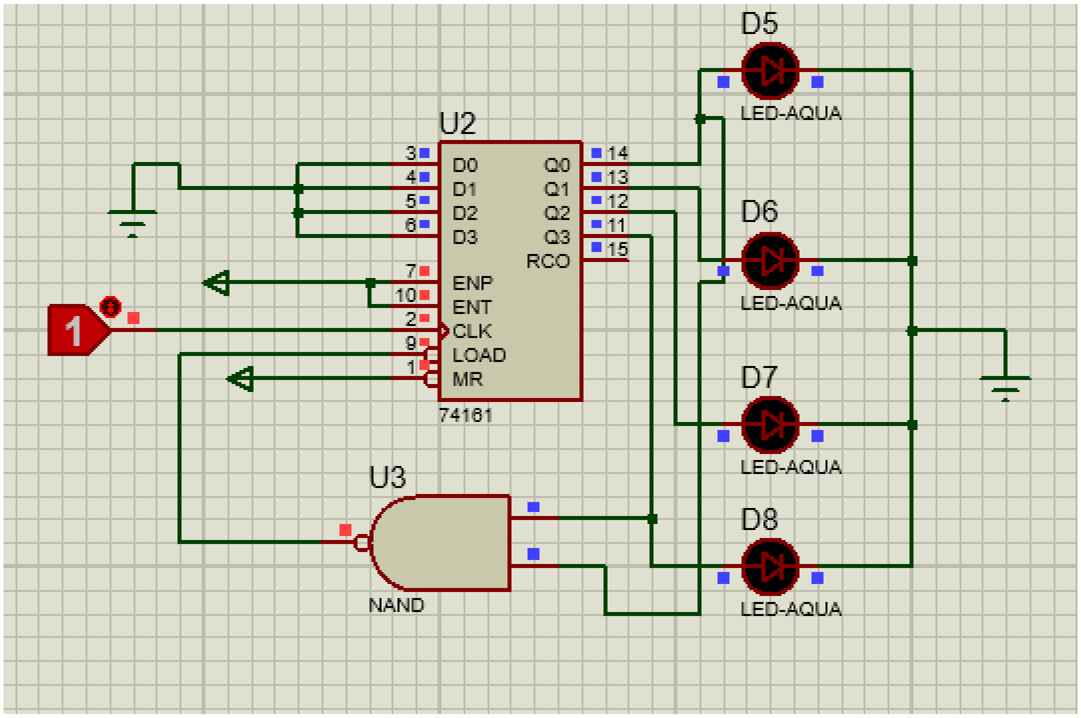

Figure:-Simulation showing the 4-bit BCD counter at 1010_2(Decimal 10) with all output LEDs in the OFF state.(When the Counter is Resetting)

Discussion:-

The experiment showed that the 74161 IC provides precise timing because all flip-flops share the same clock signal, avoiding the propagation delays found in asynchronous counters. By connecting a NAND gate to the outputs, we successfully created a BCD counter that resets after reaching 9 (1001_2).

Conclusion:-

In conclusion, the 74161 IC was successfully used to implement both 4-bit binary and BCD counters. The results verified that synchronous circuits are more reliable for digital systems due to their coordinated clocking. By using external logic to limit the count, we proved that these counters are highly adaptable for specific practical tasks.

Proteus File Google Drive link:

https://drive.google.com/drive/u/1/folders/1YrD5sh-EF4YDflSvmUrOeKAe37W3I_N2

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Department of Electrical and Electronic Engineering
Hardware / Software(✔) Report-3 Submission

Semester: Spring 26
Course Code: EEE283L
Course Title: Digital Logic Design
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Experiment No:04

Experiment Name: Study and Application of Combinational Circuit Design Blocks: Encoder, Decoder and Data Converter

Group No: 02

Group members:

Sl.	***Name***	***ID***
1.	Faiyaz Hasin Reza	24121374

Date of Submission:29/03/2026

Prepared by:

Faiyaz Hasin Reza

<u>**Experiment Name:**</u> **Study and Application of Combinational Circuit Design Blocks: Encoder, Decoder and Data Converter**

Introduction: Combinational logic circuits are the digital circuits in which the output depends only on the current input conditions. This experiment is focused on decoders and data converters which is used to convert binary information into certain output lines or human-readable format. By using integrated circuits like 74138 3 to 8 decoder and 7447 BCD to 7 segment decoder complex data routing and display interfacing can be achieved in an efficient manner in digital systems.

Objective: The goal of this experiment is to learn about the functional operation of the 74138 3-8 line decoder and the 7447 BCD to 7 segment decoder/driver. The session is aimed to verify active-low logic and chip-enable feature through the simulation while investigating how we can cascade multiple IC to design higher order decoder like 4 to 16 lines. As well as interfacing the digital logic with a common-anode seven-segment display, it is aimed at visualizing binary information as decimal digits.

Software:

- Proteus & Professional
- Digital logic library components inside Proteus

Components:

- Subcircuit component
- Logic toggle
- Logic probe
- LED indicators
- 4 to 2 encoder
- Connecting wires
- AND gate
- NOT gate
- OR gate
- 74LS138 (3-to-8 binary decoder)

4 to 2 Binary Encoder:

Simulation Diagram:

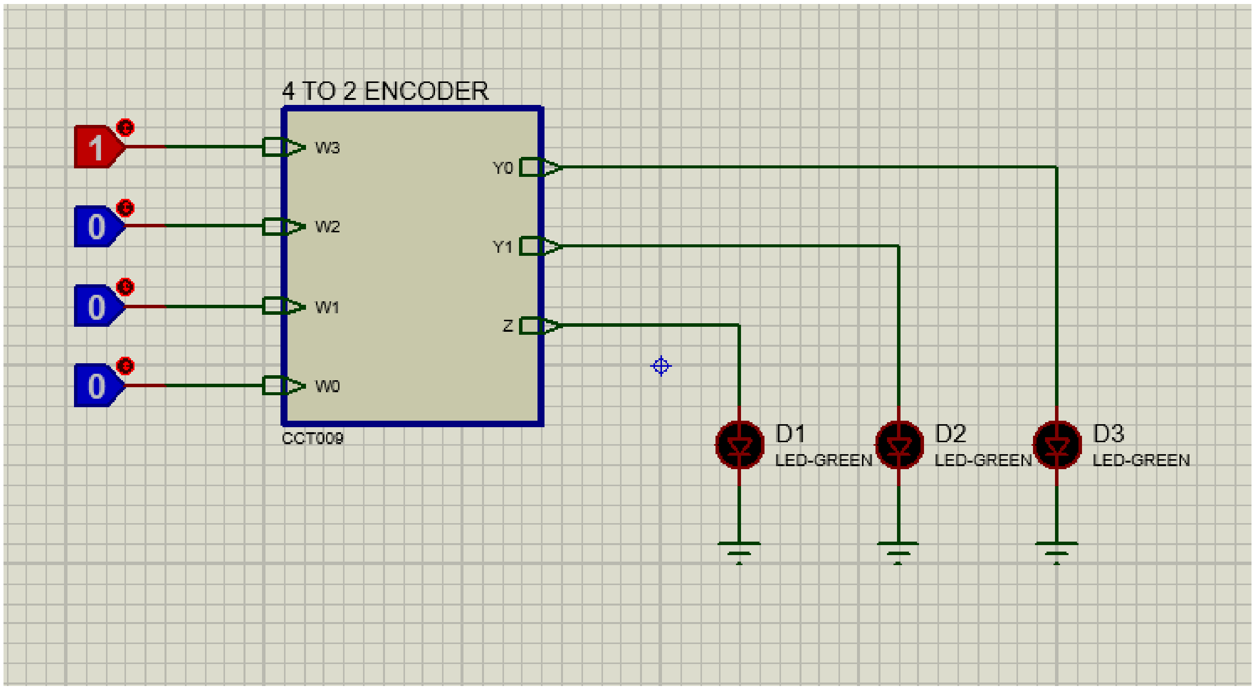

Figure:- 4 to 2 Binary Encoder circuit diagram (Parent Sheet)

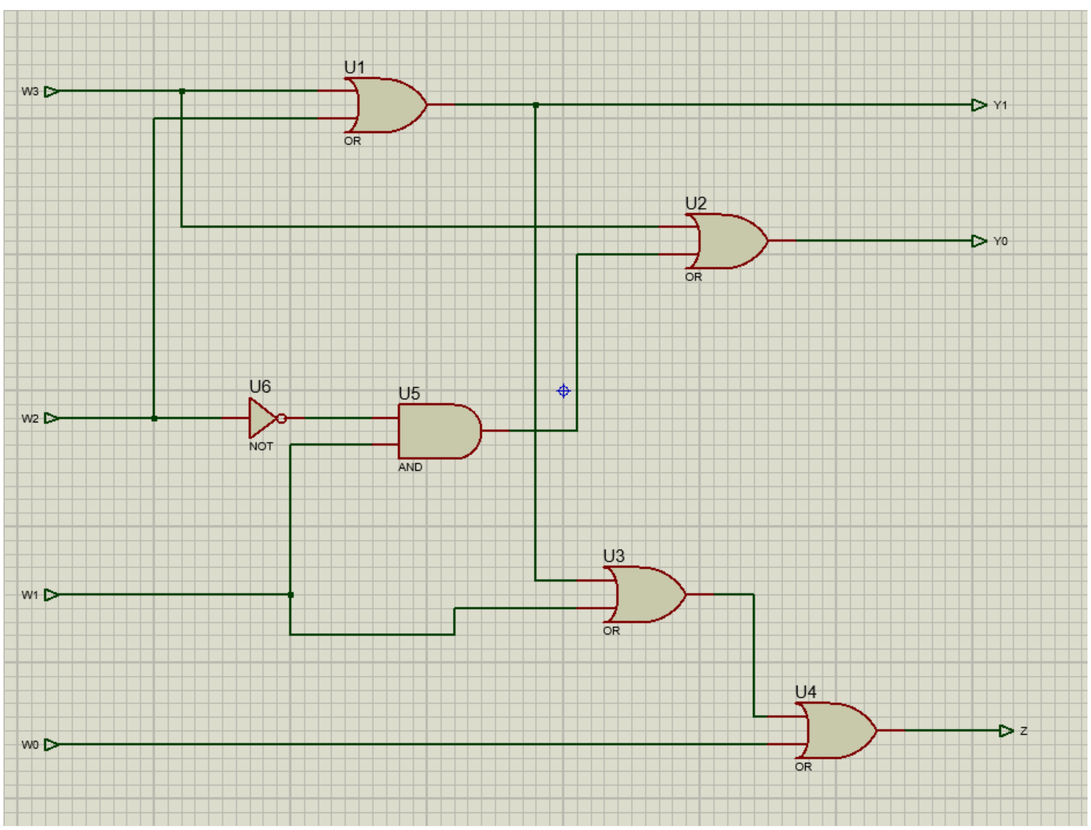

Figure:- 4 to 2 Binary Encoder circuit diagram (Child Sheet)

Truth Table:

Input				Output	
W3	**W2**	**W1**	**W0**	**Y1**	**Y0**
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1

Test Result:

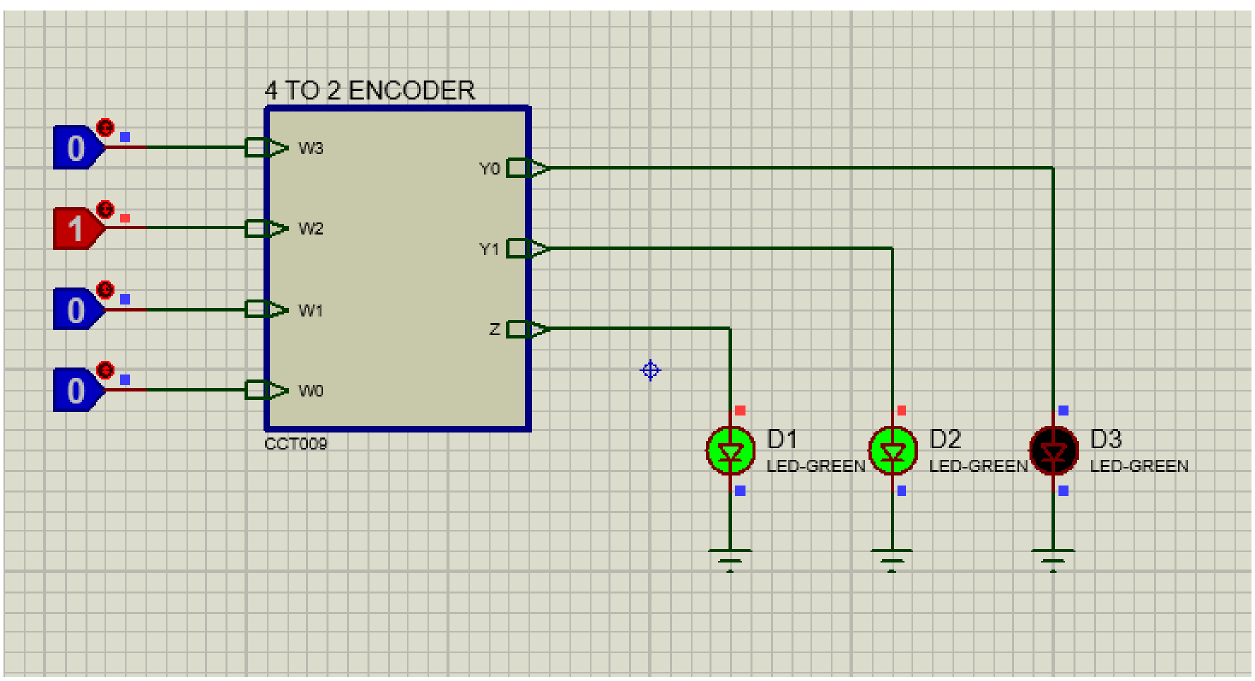

Fig: Input=W2 active, Output= Y1 selected

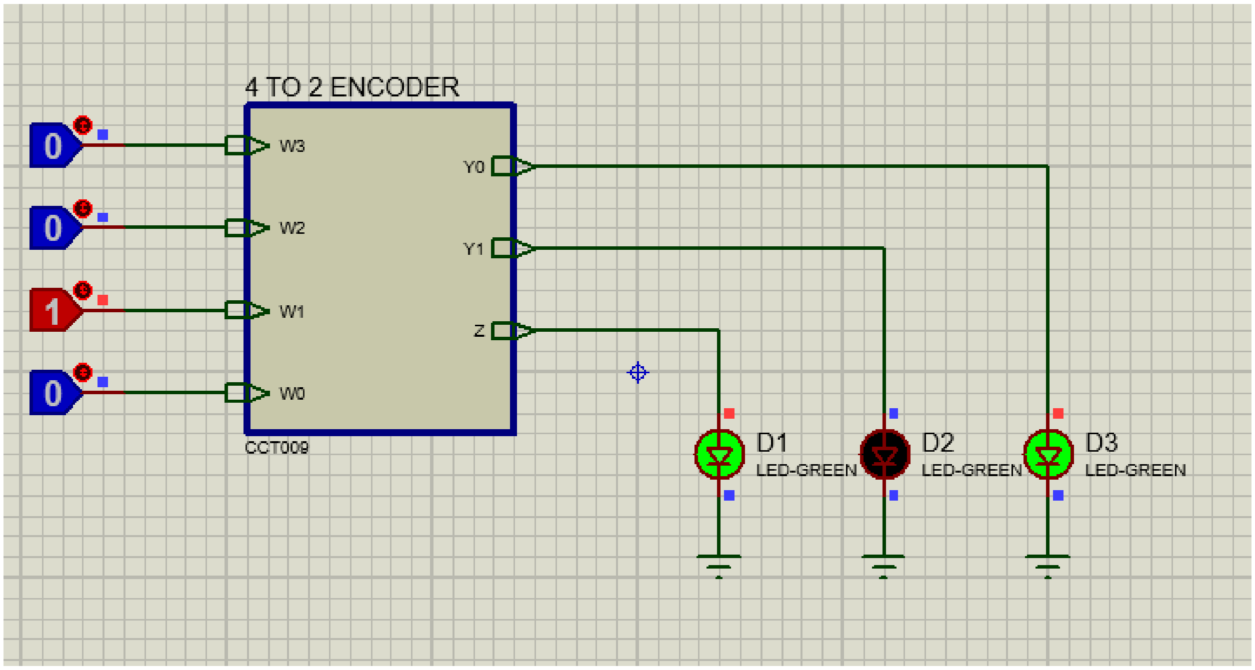

Fig: Input=W1 active, Output=Y0 selected

2 To 4 Binary Decoder:

Simulation Diagram:

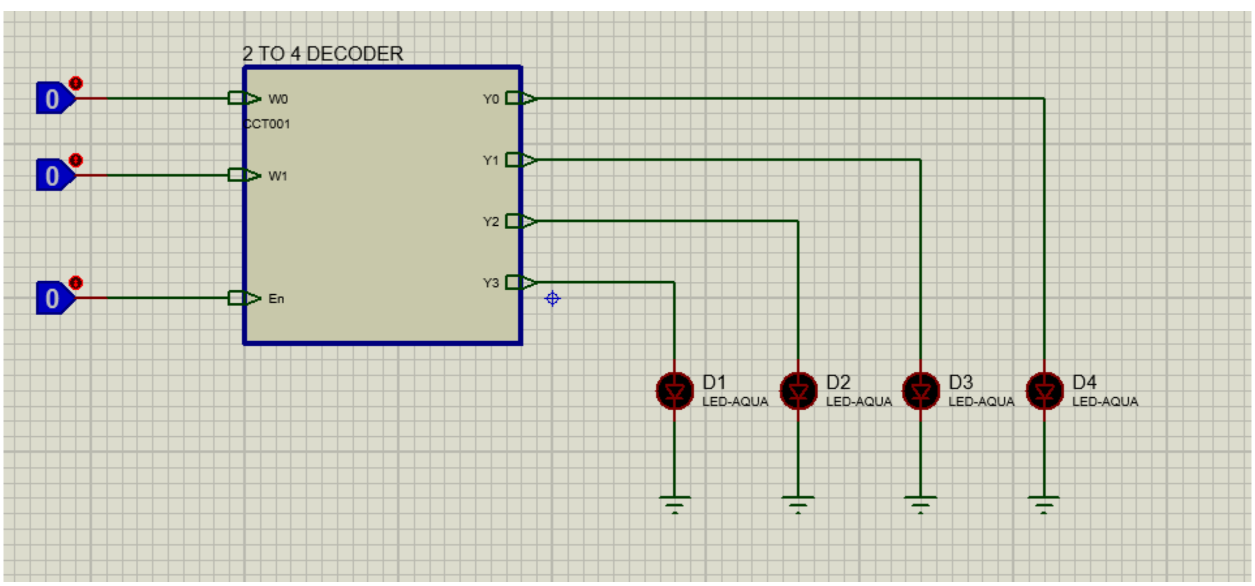

Figure:- 2 to 4 Binary Dencoder circuit diagram (Parent Sheet)

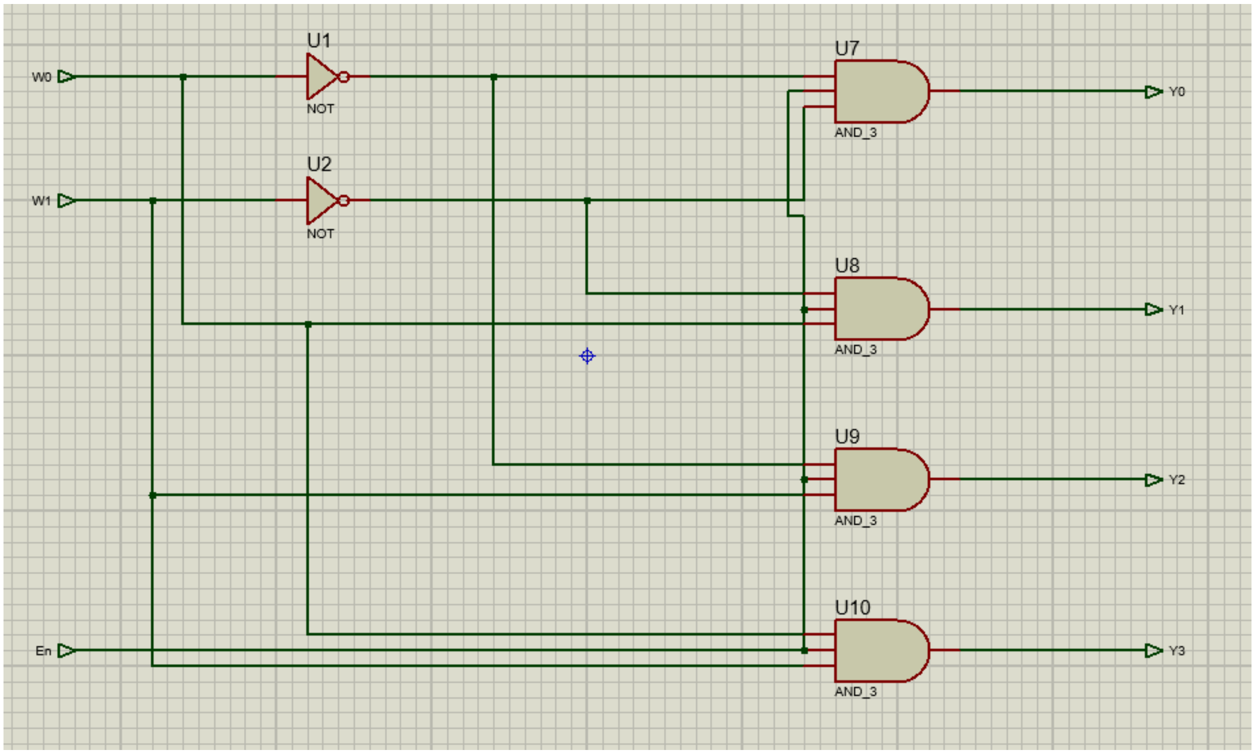

Figure:- 2 to 4 Binary Dencoder circuit diagram (Child Sheet)

Truth Table:

Input			Output			
En	**W1**	**W0**	**Y0**	**Y1**	**Y2**	**Y3**
1	0	0	1	0	0	0
1	0	1	0	1	0	0
1	1	0	0	0	1	0
1	1	1	0	0	0	1
0	x	x	0	0	0	0

Test Result:

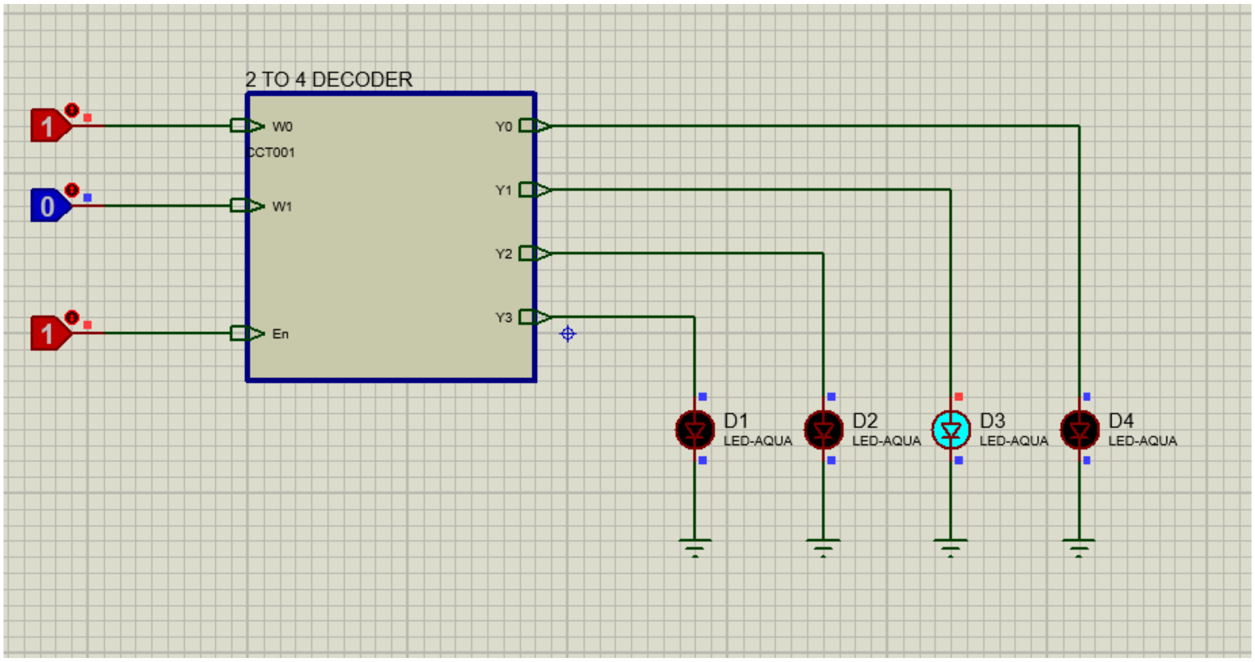

Fig: Input= En & W0 active, Output= Y1 selected

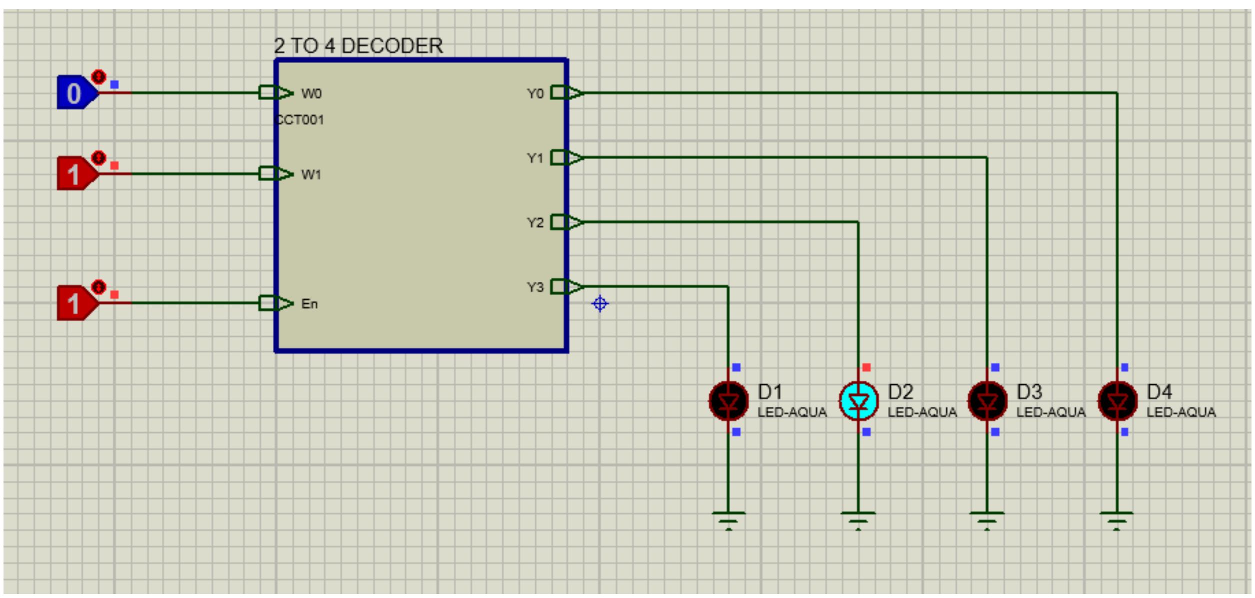

Fig: Input=En & W1 (active), Output= Y2 selected

Implement the function by using a 3-to-8 binary decoder

F(A,B,C)=$\sum m(0, 1, 3, 4, 6, 7)$

Simulation Diagram:

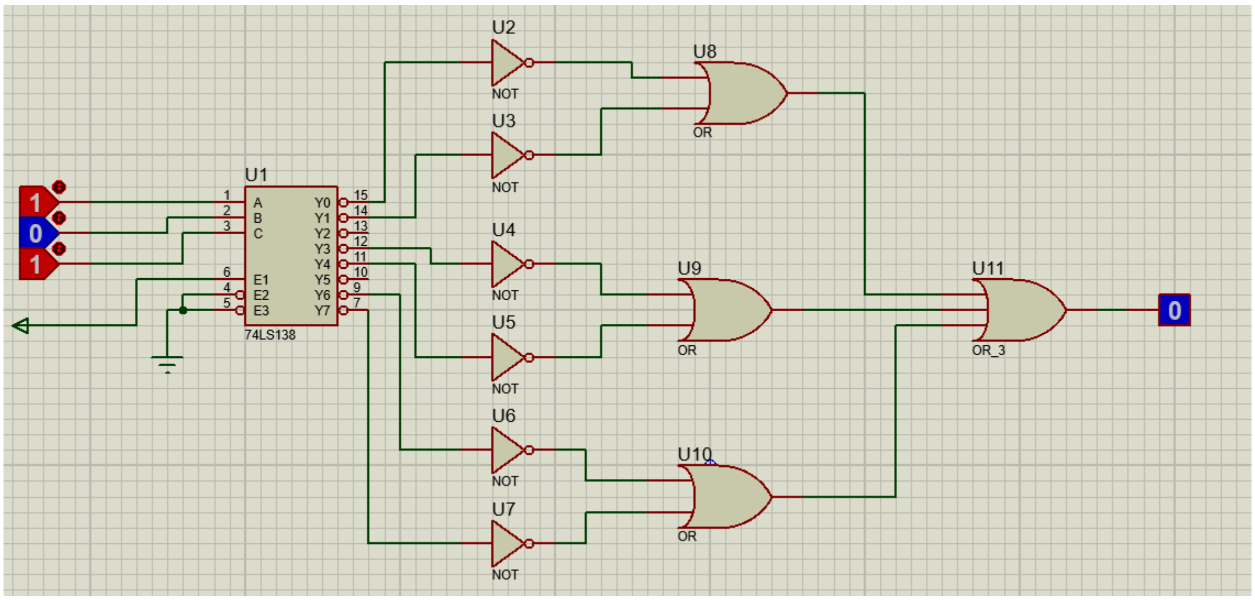

Figure:- Circuit Diagram for the function by using a 3-to-8 binary decoder

Truth Table:-

E	A	B	C	Output (F)
0	0	0	0	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	0
1	0	1	1	1
1	1	0	0	1
1	1	0	1	0
1	1	1	0	1
1	1	1	1	1

Test Result:-

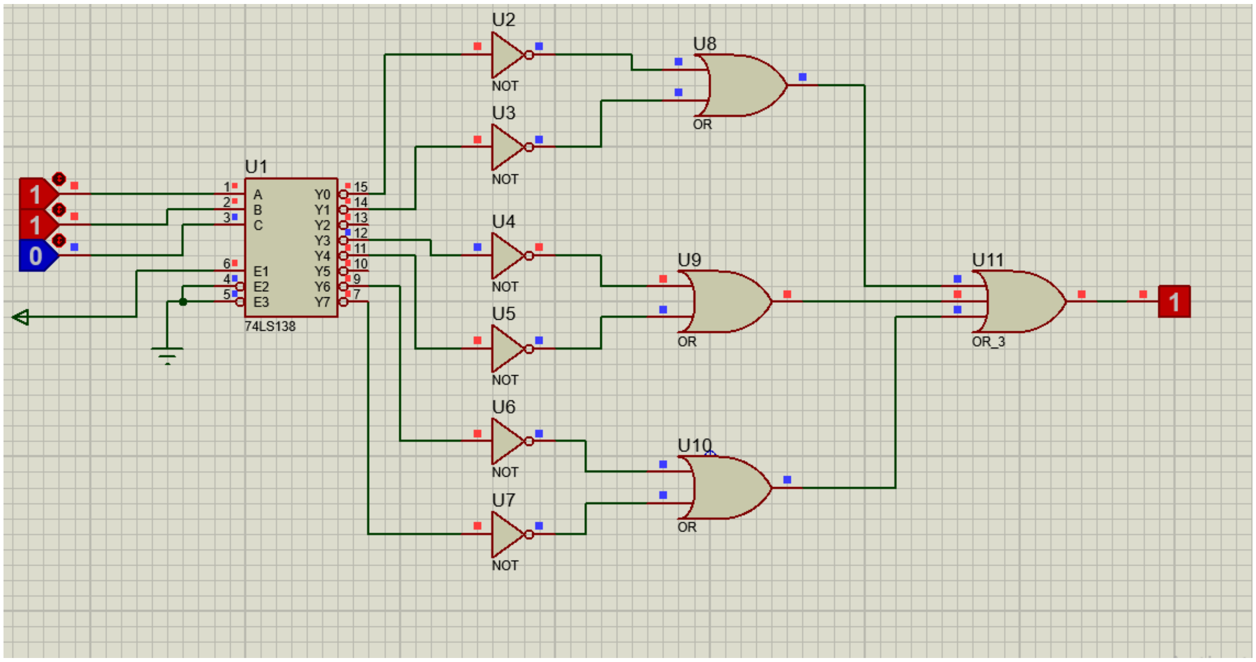

Figure:-The simulation shows the ON condition for input A=1, B=1, C=0 (Decimal 6), resulting in a logic HIGH (1) output.

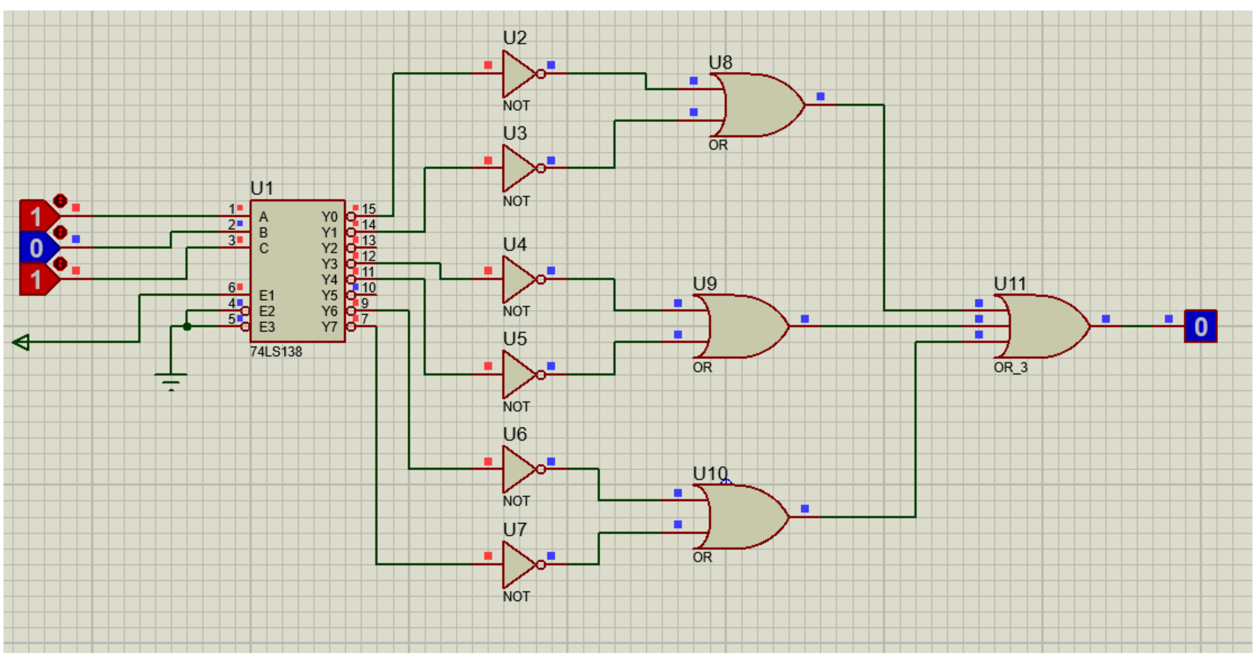

Figure:-The simulation shows the OFF condition for input A=1, B=0, C=1 (Decimal 5) resulting in a logic Low(0) output.

Discussion: The experiment successfully demonstrated that the 74138 operates on active-low logic, where the selected output is logic 0. The importance of Enable pins was observed, as they allow for circuit expansion (cascading). Finally, the 7447 IC effectively translated BCD code into the specific segment patterns required to drive a common-anode display.

Conclusion: In conclusion, this experiment provided a practical understanding of how combinational logic blocks function within a digital system. By successfully simulating the 74138 and 7447 ICs, the relationship between binary inputs and their decoded outputs was verified according to their respective truth tables. The successful implementation of a cascaded 4-to-16 decoder and a functional 7-segment display circuit confirmed the versatility of these components in data routing and visualization tasks.

Proteus File Google Drive link:

https://drive.google.com/drive/u/1/folders/1YrD5sh-EF4YDflSvmUrOeKAe37W3I_N2